G12++ Prep Programme — Data Challenge

Tasks 2–4: Analyse, Protect, Plan · Cohort 1, February 2027 · All learner data synthetic

2. Analyse

What the data tells us

60 unique learners across three NGOs after cleaning (NGO-A: 25, NGO-B: 20, NGO-C: 15). The funnel from registration to mock exam drops sharply and unevenly by NGO:

NGO	Registered	Password set	Orientation	Started topic	Completed subject	Took baseline	Took mock
NGO-A	25	24	22	17	13	21	13 (52%)
NGO-B	20	20	15	10	5	15	4 (20%)
NGO-C	15	9	7	1	1	6	0 (0%)
All	60	53	44	28	19	42	17 (28%)

- NGO-C effectively stalled: 9 of 15 learners (60%) never engaged beyond registration — no topic, no test, nothing — and zero learners reached the mock exam. This is the single largest signal in the data.
- NGO-B gets learners to the baseline well (15/20, 75%) but loses most of them before the mock (only 4/20, 20%) — a specific mid-funnel leak, not a registration problem.
- NGO-A converts best overall (13/25 reach the mock, 52%) and is the closest thing to a working funnel in this cohort.
- Device (phone vs laptop) does not explain the drop-off — 47% vs 44% of learners start a topic on each — so NGO-C's stall looks operational (onboarding, local support, connectivity), not a device/UX issue.
- For the 31 learners who sat both tests, learning gain is clearly positive: +14 to +18 percentage points on average (NGO-A: English +14.0, Maths +14.7; NGO-B: English +15.0, Maths +17.8). Real evidence the content works for learners who engage with it.

Data problems found, and how handled

- Duplicate learner row (dana.saleh@mail.example, two export snapshots with different progress) → collapsed to one row: earliest registration date, the higher value of every progress field (progress only increases), latest activity date.
- Inconsistent emails between files (e.g. " Lina.Youssef@mail.example " — padded, mixed case, in the Questionmark export) → trimmed and lower-cased before joining. Left unfixed, 3 learners' scores would have silently failed to match.
- One impossible score: huda.aziz, Mock Mathematics = 112% → dropped as a data-entry/export error; her other valid scores were kept.
- One score row for test.user@mail.example has no matching learner in the app export → dropped as a QA/test record.
- Checked explicitly for duplicate (learner, assessment, subject) score rows after normalising emails — found none, so no double-counted averages.

How confident am I

Funnel counts: high confidence — plain counts after the fixes above, covered by unit tests (tests/test_pipeline.py).

Learning gain: moderate confidence only. It is measured on the 31 of 60 learners who sat both tests — the most motivated, most-retained group. That is survivorship bias: we don't know the gain for the 43 who registered but never reached the mock, and NGO-C has zero data points at all, so "the programme works" really means "it works for the roughly half who stick with it." Per-NGO gain comparisons also rest on small samples (≤15 paired learners), so differences between NGO-A and NGO-B could be noise.

Two recommended product changes

- Close the registration → first-topic gap (53% of all learners never start a topic, and it is NGO-C's exact failure point): trigger an automatic nudge a few days after orientation if no topic has started, and add first-session telemetry (app opened, content downloaded, connectivity) so we can tell apart "didn't open the app" from "couldn't load content" — right now the data only shows the outcome, not the cause.
- Add a baseline → mock checkpoint for coordinators, not just learners, aimed first at NGO-B (75% reach baseline, only 20% reach mock — the sharpest single-stage drop in the data): a visible "on track for mock"

flag per learner so an NGO can intervene in week 3, not discover the drop-off from a quarterly export in February.

3. Protect

Every learner in this dataset is a young person, registered through an NGO across several countries, and the combined files (name, phone, DOB, NGO, device, exam scores) are enough to fully identify and locate a specific minor and know exactly how they performed. That should be treated as sensitive children's data by default, regardless of which country's specific law applies.

- Minimise: analytics and dashboards (including the one built for this test) should run on a pseudonymous learner ID, never on full_name / phone / date_of_birth directly — those fields belong only in the app's own learner-profile store, not in any export used for reporting or analysis.
- Scope access: an NGO should only ever see its own learners (role-based access scoped by the ngo field); individual learner drill-down should be restricted to the coordinator role, with funders/leadership seeing NGO- and cohort-level aggregates only.
- Control the pipeline: routine flat-file CSV handoffs between the app, Questionmark and NGOs (like the ones used for this test) are themselves a privacy risk — replace with an access-controlled, logged pipeline/API, and treat any CSV that leaves that pipeline as a reportable event.
- Vendor risk: Questionmark is a data processor — confirm a data processing agreement restricting them from retaining or reusing learner data beyond returning scores, and apply the same standard to any analytics/BI tool used downstream.
- Cross-border transfer: confirm where learner data is hosted and processed against each partner country's child-data rules, and prefer regional data residency where that's required rather than one global store by default.
- Retention: define and enforce a retention window for raw PII (e.g. deleted a fixed period after a learner exits the programme), keeping only pseudonymised, aggregated metrics for longer-term reporting; give NGOs a deletion-request path on behalf of a learner or guardian.

4. Plan — growing from 150 to 750 participants in 12 months

Priorities

- Fix what is already broken before multiplying it: NGO-C's stall and NGO-B's baseline→mock leak are operational problems visible today at 150 learners — scaling 5× without fixing them scales the failure, not the programme.
- Instrument the funnel in near-real-time, per NGO, so a coordinator can act in week 3, not find out from a quarterly CSV export.
- Turn what works at NGO-A (the best-converting funnel in the data) into a repeatable onboarding playbook for new NGO partners, instead of each NGO re-inventing its own process.
- Productise the funnel + learning-gain view built for this test into a real NGO-facing reporting feature, rather than a one-off analysis artifact.

What I would not do

- Not onboard new NGOs or learners ahead of shipping the funnel fixes and real-time reporting above — adding scale to a broken funnel multiplies drop-off, not outcomes.
- Not build a general-purpose BI/analytics platform — keep reporting scoped to the funnel and learning-gain views NGOs have actually needed so far; no speculative dashboards nobody asked for.
- Not keep manual/periodic CSV exports as the Questionmark↔app integration at 5× scale and more countries — it will not hold up operationally or for privacy (see Task 3).

How I would measure success

- Primary: % of registered learners reaching the mock exam, per NGO, tracked monthly — today 28% overall and 0% for NGO-C.
- Secondary: average learning gain among mock-takers, and the share of registered learners who become mock-takers with a valid paired score — directly closing the survivorship-bias gap identified in Task 2.
- Leading indicator: time from registration to first topic started, per NGO — detects an NGO-C-style stall weeks before the mock-exam outcome would reveal it.

Testing whether NGOs would pay for reporting

- Ship the funnel + learning-gain dashboard built for this test as a free, lightweight feature at the current 150-learner scale, instrumented to see whether coordinators actually open it and act on it.
- Run a willingness-to-pay pilot with one or two NGOs: offer a paid tier for features they ask for directly (export/API access, a weekly digest, individual-learner drill-down) and see whether they adopt or stall on it.
- Treat unprompted use — a coordinator checking the dashboard weekly and raising an NGO-C-style stall with us before we spot it — as the real signal of willingness to pay; a feature nobody opens will not convert, whatever an interview suggests.